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BIO 101: Introduction to Biology - Cells, Genetics and Diversity

The organising ideas of life, from the cell to ecosystems

FACULTY PRIMER

Faculty Primer

Faculty of Bio-Sciences
Botany | 100L

Written for students, by students.

Practical notes, worked examples, practice questions and a revision plan.

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Independent study support | Not an official UNIZIK publication

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Why this pack exists

BIO 101 covers an enormous syllabus, so the danger is memorising isolated facts. This pack organises the course around four unifying ideas - cells, information, energy and evolution - and gives you diagram-and-label practice for the questions that always appear.

This is a student-written study companion, not a reproduced lecture handout or a leaked examination paper. Use it to organise the topics you are taught, practise the kind of thinking your course rewards, and walk into class with better questions.

What you will be able to do

- Compare prokaryotic and eukaryotic cells and describe the functions of major organelles.
- Explain the cell cycle, mitosis and meiosis and why the distinction matters.
- Apply Mendelian genetics to simple crosses and interpret pedigree information.
- Classify organisms using binomial nomenclature and explain the basis of biological diversity.

How to use this pack

1. Skim the whole pack once in 20 minutes so you know what is inside.
2. Work through one section at a time with your lecture note and recommended text beside you.
3. Attempt the practice questions before reading the answers. Mark what you missed.
4. Use the final checklist to plan your revision week, one tick per study session.

> **Free to download, always**
> This pack is free. There is no fee, no subscription and no payment step before download, and the full PDF is not watermarked or locked. Share the link with a coursemate instead of forwarding the file.

> **Confirm your official course outline**
> Course codes, credit units and outlines can differ by department and session. Treat this pack as supplementary support and confirm the current outline with your lecturer, course rep or departmental handbook before an assessment.

Cells as the unit of life

Every living thing is made of cells, and every cell maintains a boundary, processes energy and stores information.

- Prokaryotes lack a membrane-bound nucleus; eukaryotes have one.
- Nucleus: stores DNA and coordinates gene expression.
- Mitochondria: site of aerobic respiration and ATP production.
- Chloroplasts: photosynthesis in plants and algae.
- Ribosomes: protein synthesis; rough endoplasmic reticulum folds proteins.

- Cell membrane: selectively permeable phospholipid bilayer controlling exchange.

> **Label diagrams in the exam**
> Draw large, label with straight lines, and give a function beside each label. Marks are usually split between identification and function.

Cell division and inheritance

Mitosis produces identical cells for growth; meiosis produces gametes with half the chromosome number.

- Mitosis: one division, two diploid daughter cells, genetically identical.
- Meiosis: two divisions, four haploid cells, genetic variation through crossing over and independent assortment.
- Genotype is the genetic makeup; phenotype is the observable trait.
- A dominant allele is expressed when present; a recessive allele is expressed only when homozygous.
- Use a Punnett square for monohybrid crosses and state ratios clearly.
- Pedigrees can indicate whether a trait is dominant, recessive, autosomal or sex-linked.

Energy, ecology and diversity

Life persists by capturing energy and cycling matter through ecosystems.

- Photosynthesis converts light energy into chemical energy; respiration releases it.
- Food chains show energy flow; only about 10 percent passes to the next trophic level.
- Nutrient cycles such as carbon, nitrogen and water connect living and non-living systems.
- Binomial nomenclature: genus capitalised, species lowercase, both italicised.
- Classification hierarchy: domain, kingdom, phylum, class, order, family, genus, species.
- Variation plus selection is the mechanism behind adaptation and speciation.

The levels of organisation

Biology studies living systems at every scale: molecules, organelles, cells, tissues, organs, organisms, populations, communities and ecosystems. The same question - how is this organised and how does it maintain itself - gets a different answer at each level, and part of learning the subject is knowing which level a question belongs to.

The unifying ideas are few and powerful: cells are the basic unit of life; all living things use the same molecular machinery (DNA, RNA, proteins, ATP); structure and function are related at every level; and evolution by natural selection explains both diversity and similarity.

A cell is not a bag. It is compartmentalised: the membrane regulates exchange, the nucleus holds genetic material in eukaryotes, mitochondria generate ATP, ribosomes make protein, and the endomembrane system moves material. Prokaryotes lack membrane-bound organelles and are nonetheless biochemically sophisticated.